

Question 1

Case	Activity	Resource	Event type	Timestamp		Case	Activity	Resource	Event type	Timestamp
1	Check Stock	Ali	Offer	01/01/2021 09:00		2	Get Raw Materials From Supplier 1	Osama	Allocate	01/01/2021 12:15
1	Check Stock	Ahmed	Offer	01/01/2021 09:00		2	Get Raw Materials From Supplier 1	Osama	Start	01/01/2021 12:15
1	Check Stock	Ali	Allocate	01/01/2021 09:15		2	Get Raw Materials From Supplier 1	Osama	Complete	01/01/2021 14:15
1	Check Stock	Ali	Start	01/01/2021 09:30		2	Manufacture Product	Ali	Offer	01/01/2021 14:15
1	Check Stock	Ali	Complete	01/01/2021 11:00		2	Manufacture Product	Ali	Allocate	01/01/2021 14:30
2	Check Stock	Ahmed	Offer	01/01/2021 09:15		2	Manufacture Product	Ali	Start	01/01/2021 14:30
2	Check Stock	Ahmed	Allocate	01/01/2021 09:30		2	Manufacture Product	Ali	Suspend	01/01/2021 16:30
2	Check Stock	Ahmed	Start	01/01/2021 09:30		2	Manufacture Product	Ali	Start	01/01/2021 17:30
2	Check Stock	Ahmed	Complete	01/01/2021 11:00		2	Manufacture Product	Ahmed	Allocate	02/01/2021 09:30
1	Retrieve Product	Osama	Offer	01/01/2021 11:15		2	Manufacture Product	Ahmed	Start	02/01/2021 09:30
1	Retrieve Product	Ahmed	Offer	01/01/2021 11:15		2	Manufacture Product	Ahmed	Suspend	02/01/2021 10:30
1	Retrieve Product	Ahmed	Allocate	01/01/2021 11:30		2	Manufacture Product	Ahmed	Start	02/01/2021 12:00
1	Retrieve Product	Ahmed	Start	01/01/2021 11:30		2	Manufacture Product	Ahmed	Complete	02/01/2021 14:30
1	Retrieve Product	Ahmed	Suspend	01/01/2021 13:00		2	Ship Product	Ahmed	Offer	02/01/2021 14:30
1	Retrieve Product	Ahmed	Start	01/01/2021 13:30		2	Ship Product	Ali	Offer	02/01/2021 14:30
1	Retrieve Product	Ahmed	Complete	01/01/2021 14:30		2	Ship Product	Ahmed	Allocate	02/01/2021 14:45
2	Check Raw	Osama	Offer	01/01/2021 11:30		2	Ship Product	Ahmed	Start	02/01/2021 15:00
2	Check Raw	Osama	Allocate	01/01/2021 11:30		2	Ship Product	Ahmed	Suspend	02/01/2021 16:00
2	Check Raw	Osama	Start	01/01/2021 11:30		2	Ship Product	Ahmed	Start	02/01/2021 17:30
2	Check Raw	Osama	Complete	01/01/2021 12:30		2	Ship Product	Ahmed	Complete	02/01/2021 19:30
1	Ship Product	Ahmed	Offer	01/01/2021 14:30		1	Receive Payment	Ali	Offer	02/01/2021 15:15
1	Ship Product	Ali	Offer	01/01/2021 14:30		1	Receive Payment	Ali	Allocate	02/01/2021 15:30
1	Ship Product	Ahmed	Allocate	01/01/2021 14:45		1	Receive Payment	Ali	Start	02/01/2021 16:00
1	Ship Product	Ahmed	Start	01/01/2021 14:45		1	Receive Payment	Ali	Complete	02/01/2021 17:00
1	Ship Product	Ahmed	Suspend	01/01/2021 16:15		2	Receive Payment	Ali	Offer	03/01/2021 09:00
1	Ship Product	Ahmed	Start	01/01/2021 16:45		2	Receive Payment	Ali	Allocate	03/01/2021 09:30
1	Ship Product	Ahmed	Complete	01/01/2021 17:45		2	Receive Payment	Ali	Start	03/01/2021 10:00
2	Get Raw Materials From Supplier 1	Osama	Offered	01/01/2021 12:00		2	Receive Payment	Ali	Complete	03/01/2021 11:00

1. Effective time (in hours) for resource “Ali” within activity “Manufacture Product” in case 2.

$(16:30 - 14:30) + (9:30 \text{ 1/1/2021} - 17:30 \text{ 2/1/2021}) = 18 \text{ hours}$

2. Service time (in hours) for resource “Ali” within activity “Check Stock” in case 1.

$11:00 - 9:15 = 1 \text{ hour and } 45 \text{ minutes}$

3. Choose the correct Sojourn time (in hours) and minutes for activity “Ship product” in case 2.

$19:30 - 14:30 = 5 \text{ hours}$

Question 2

BP Case	Activity	Resource	Event Type	Timestamp
Case 1	Receive request MAS	Doaa	Offered	10 –05-2024 9:15
Case 1	Receive request MAS	Doaa	Started	10 –05-2024 9:45
Case 1	Receive request MAS	Doaa	Suspended	10 –05-2024 11:45
Case 2	Receive request MAS	Nour	Allocated	10 –05-2024 11:45
Case 2	Receive request MAS	Nour	Started	10 –05-2024 12:45
Case 2	Receive request MAS	Nour	Completed	10 –05-2024 16:45
Case 2	Receive request MAS	Doaa	Started	10 –05-2024 17:45
Case 2	Receive request MAS	Doaa	Completed	10 –05-2024 20:45
Case 2	Assign date for MAS	Nour	Offered	11 –05-2024 08:15
Case 2	Assign date for MAS	Nour	Allocated	11 –05-2024 09:15
Case 2	Assign date for MAS	Nour	Started	12 –05-2024 10:15
Case 2	Assign date for MAS	Nour	Completed	12 –05-2024 13:15

1. Choose the correct waiting time (in hours) for within activity “Receive request MAS” in case 1.

9:45 – 9:15 = 30 minutes

2. Choose the correct effective time (in hours) for resource “NOUR” within activity “Receive request MAS” in case 2.

16:45 – 12:45 = 4 hours

Question 3

BP Case	Activity	Resource	Event Type	Timestamp
Case 1	Prepare Samples	Maha	Offered	10 –05-2024 11:30
Case 1	Prepare Samples	Maha	Started	10 –05-2024 12:30
Case 1	Write Report	Dina	Offered	10 –05-2024 16:30
Case 1	Prepare Samples	Maha	Suspended	10 –05-2024 17:30
Case 1	Prepare Samples	Maha	Started	10 –05-2024 18:30
Case 1	Prepare Samples	Dina	Suspended	10 –05-2024 19:30
Case 1	Write Report	Dina	Allocated	10 –05-2024 20:30
Case 1	Write Report	Dina	Started	10 –05-2024 22:30
Case 1	Write Report	Dina	Completed	10 –05-2024 23:30
Case 1	Prepare Samples	Salma	Started	11 –05-2024 14:30
Case 1	Prepare Samples	Salma	Completed	11 –05-2024 20:30

1. Choose the correct Service time (in hours) and minutes for resource “Dina” within activity “Write Report” in case 1
 $23:30 - 20:30 = 3 \text{ hours}$
2. Choose the correct Sojourn time (in hours) and minutes for activity “Prepare Samples” in case 1
 $20:30 \text{ 11/5/2024} - 11:30 \text{ 10/5/2024} = 12 \text{ hours} + 30 \text{ minutes} + 20 \text{ hours} + 30 \text{ minutes} = 33 \text{ hours} +$

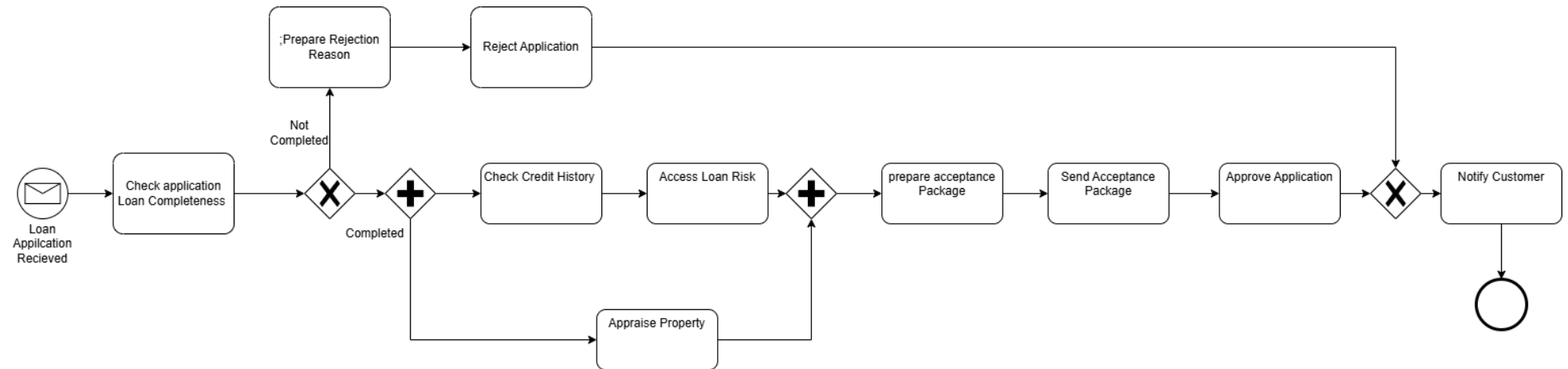
Question 4

Case ID	Task	Resource	Event type	Timestamp
1	A	R1	Start	03-01-2021 09:00:00
1	B	R1	Start	03-01-2021 10:00:00
1	A	R1	Complete	03-01-2021 11:00:00
1	C	R1	Start	03-01-2021 11:30:00
1	B	R1	Complete	03-01-2021 12:30:00
1	C	R1	Complete	03-01-2021 13:30:00

If there is an error in the log recording, then the most suitable resolution technique to resolve events overlapping is:

Defer Low priority item (DL).

Question 5



Given the following details:

- Check application Loan Completeness takes 5 hours
- Preparing Rejection Reason takes 2 hours
- Application rejection takes 1.5 hours
- Checking Credit History and Appraising Property takes 3 hours
- Accessing Loan Risk takes 0.5 hours
- preparing acceptance Package takes 1 day = 24 hours
- Sending Acceptance Package takes 2 hours
- Approving Application takes 1.5 hours
- Notifying the Customer takes 0.5 hour.

In Case 1, the Application is completed by the customer, while in Case 2, the application is not completed and the bank needs more info from the customer. Calculate the effective time per case. (Only Write The number of hours, For Example: 3)

1. Effective time for Case 1 in hours

$$5 + (\text{MAX}[(3 + 0.5), 3] + 24 + 2 + 1.5) + 0.5 = 36.5 \text{ hours}$$

2. Effective time for Case 2 in hours

$$5 + 3.5 + 0.5 = 9 \text{ hours}$$